MTH 166

Lecture-30

Gradient of a Scalar field

Topic:

Vector Differential Calculus

Learning Outcomes:

1. To calculate gradient of a scalar field

2. To find normal vector and unit normal vector to a surface

3. To find angle between two given surfaces

Gradient of a scalar field

Let $f(x, y, z)$ be a given scalar surface.

Let us consider a vector differential operator called as **Del/Nabla** defined as:

$$\vec{\nabla} = \left(\hat{\imath}\frac{\partial}{\partial x} + \hat{\jmath}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right)$$

Then gradient of scalar field f is given by:

$$\text{grad}(f) = \vec{\nabla}f = \left(\hat{\imath}\frac{\partial}{\partial x} + \hat{\jmath}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right)f(x, y, z)$$

$$= \left(\hat{\imath}\frac{\partial f}{\partial x} + \hat{\jmath}\frac{\partial f}{\partial y} + \hat{k}\frac{\partial f}{\partial z}\right)$$

$$= \left(f_x\hat{\imath} + f_y\hat{\jmath} + f_z\hat{k}\right).$$

Problem 1. Compute the gradient of function $(x^3 - 3x^2y^2 + y^3)$ at point (1,2).

Solution. Let $f(x, y, z) = (x^3 - 3x^2y^2 + y^3)$

$\Rightarrow f_x = (3x^2 - 6xy^2), f_y = (3y^2 - 6x^2y)$

Then gradient of scalar field f is given by:

$\text{grad}(f) = \vec{\nabla}f = \left(f_x\hat{\imath} + f_y\hat{\jmath}\right) = (3x^2 - 6xy^2)\hat{\imath} + (3y^2 - 6x^2y)\hat{\jmath}$

At point (1,2):

$\text{grad}(f) = \vec{\nabla}f = (3 - 24)\hat{\imath} + (12 - 12)\hat{\jmath} = -21\hat{\imath}$ **Answer.**

Problem 2. Compute the gradient of function $\log(x+y+z)$ at point $(1,2,-1)$.

Solution. Let $f(x,y,z) = \log(x+y+z)$

$$\Rightarrow f_x = f_y = f_z = \frac{1}{(x+y+z)}$$

Then gradient of scalar field f is given by:

$$\operatorname{grad}(f) = \vec{\nabla} f = \left(f_x\hat{\imath} + f_y\hat{\jmath} + f_z\hat{k}\right) = \frac{1}{(x+y+z)}\left(\hat{\imath} + \hat{\jmath} + \hat{k}\right)$$

At point $(1,2.-1)$:

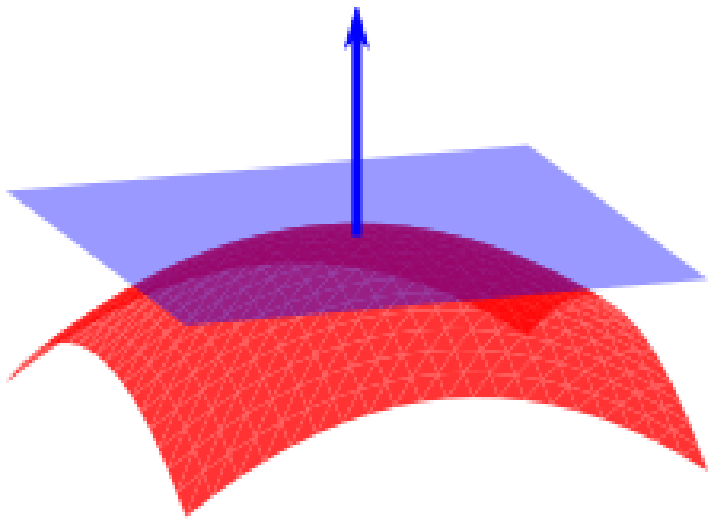
$$\operatorname{grad}(f) = \vec{\nabla} f = \frac{1}{(1+2-1)}\left(\hat{\imath} + \hat{\jmath} + \hat{k}\right) = \frac{1}{2}\left(\hat{\imath} + \hat{\jmath} + \hat{k}\right)$$ **Answer.**

Normal Vector to a Scalar Field Surface

Geometrically, Gradient of a scalar field represents a vector normal to the surface,

Let $f(x, y, z)$ be a given scalar surface.

Let $\vec{\nabla} = \left(\hat{\imath}\frac{\partial}{\partial x} + \hat{\jmath}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right)$



Then vector normal to the surface f is given by:

Normal vector, $\vec{n} = \mathrm{grad}(f) = \vec{\nabla} f = \left(f_x\hat{\imath} + f_y\hat{\jmath} + f_z\,\hat{k}\right)$

Unit normal vector, $\hat{n} = \frac{\vec{n}}{|\vec{n}|} = \frac{\vec{\nabla} f}{|\vec{\nabla} f|}$

Problem 1. Find the vector normal and unit normal to the surface $y^2 = 16x$ at (4,8).

Solution. Let $f(x, y, z) = 16x - y^2$

$\Rightarrow f_x = 16, \quad f_y = -2y$

Then gradient of scalar field f is given by:

$$\text{grad}(f) = \vec{\nabla} f = \left(f_x \hat{\imath} + f_y \hat{\jmath}\right) = 16\hat{\imath} - 2y\hat{\jmath}$$

At point (4,8): $\vec{\nabla} f = 16\hat{\imath} - 16\hat{\jmath}$

Normal vector, $\vec{n} = \vec{\nabla} f = 16\hat{\imath} - 16\hat{\jmath}$

Unit normal vector, $\hat{n} = \frac{\vec{n}}{|\vec{n}|} = \frac{\vec{\nabla} f}{|\vec{\nabla} f|} = \frac{16\hat{\imath} - 16\hat{\jmath}}{\sqrt{(16)^2 + (16)^2}} = \frac{16(\hat{\imath} - \hat{\jmath})}{16\sqrt{1+1}} = \frac{(\hat{\imath} - \hat{\jmath})}{\sqrt{2}}$ **Answer.**

Problem 2. Find vector normal and unit normal to the surface $x^2 + y^2 + z^2 = 4$ at $(1,1,1)$.

Solution. Let $f(x, y, z) = x^2 + y^2 + z^2 - 4$

$\Rightarrow f_x = 2x, \quad f_y = 2y, \qquad f_z = 2z$

Then gradient of scalar field f is given by:

$\text{grad}(f) = \vec{\nabla} f = \left(f_x\hat{\imath} + f_y\hat{\jmath} + f_z\hat{k}\right) = 2x\hat{\imath} + 2y\hat{\jmath} + 2z\hat{k}$

At point $(1,1,1)$: $\vec{\nabla} f = 2\hat{\imath} + 2\hat{\jmath} + 2\hat{k}$

Normal vector, $\vec{n} = \vec{\nabla} f = 2\hat{\imath} + 2\hat{\jmath} + 2\hat{k}$

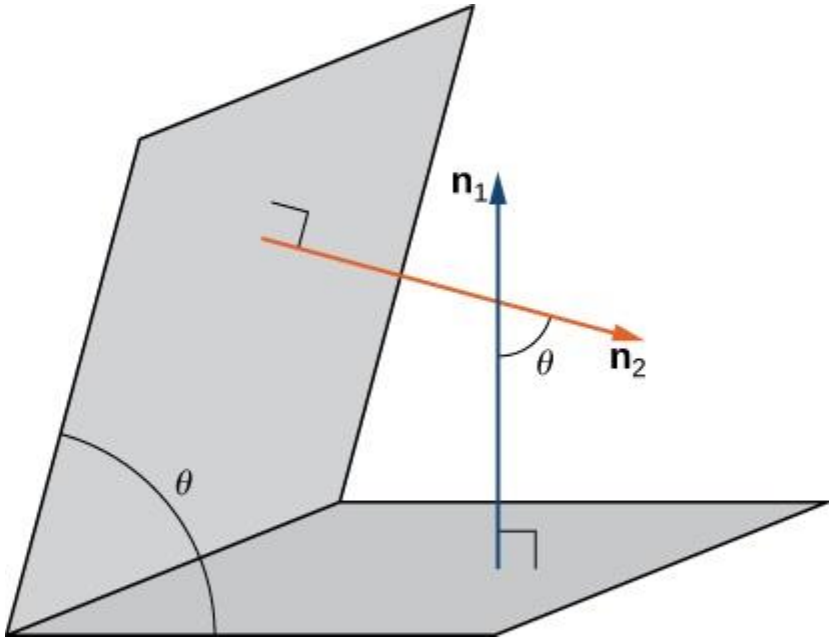
Unit normal vector, $\hat{n} = \frac{\vec{n}}{|\vec{n}|} = \frac{\vec{\nabla} f}{|\vec{\nabla} f|} = \frac{2\hat{\imath}+2\hat{\jmath}+2\hat{k}}{\sqrt{4+4+4}} = \frac{2(\hat{\imath}+\hat{\jmath}+\hat{k})}{2\sqrt{3}} = \frac{(\hat{\imath}+\hat{\jmath}+\hat{k})}{\sqrt{3}}$ **Answer.**

Angle between two Scalar Surface

Angle between two surfaces is equal to the angle between their normal.

Let f and g be two given scalar surfaces. Let $\overrightarrow{n_1}$ and $\overrightarrow{n_2}$ be the vectors normal to the surfaces f and g respectively.

Then angle between surfaces f and g is given by:



$$\cos\theta = [\widehat{n_1}.\widehat{n_2}] = \left[\frac{\overrightarrow{n_1}}{|\overrightarrow{n_1}|}.\frac{\overrightarrow{n_2}}{|\overrightarrow{n_2}|}\right]$$

$$\Rightarrow \theta = cos^{-1}\left[\frac{\vec{\nabla}f}{|\vec{\nabla}f|}.\frac{\vec{\nabla}g}{|\vec{\nabla}g|}\right]$$

Problem. Find the angle between the surfaces $z = x^2 + y^2, z = 2x^2 - 3y^2$ at (2,1,5).

Solution. Let $f(x, y, z) = x^2 + y^2 - z$

$\Rightarrow f_x = 2x, \quad f_y = 2y, \quad f_z = -1$

Then gradient of scalar field f is given by:

$$\text{grad}(f) = \vec{\nabla} f = \left(f_x \hat{\imath} + f_y \hat{\jmath} + f_z \hat{k}\right) = 2x\hat{\imath} + 2y\hat{\jmath} - \hat{k}$$

At point (2,1,5): $\vec{\nabla} f = 4\hat{\imath} + 2\hat{\jmath} - \hat{k}$

Normal vector, $\overrightarrow{n_1} = \vec{\nabla} f = 4\hat{\imath} + 2\hat{\jmath} - \hat{k}$

Let $g(x, y, z) = 2x^2 - 3y^2 - z$

$\Rightarrow g_x = 4x, \quad g_y = -6y, \quad g_z = -1$

Then gradient of scalar field g is given by:

$$\text{grad}(g) = \vec{\nabla} g = \left(g_x\hat{\imath} + g_y\hat{\jmath} + g_z\hat{k}\right) = 4x\hat{\imath} - 6y\hat{\jmath} - \hat{k}$$

At point (2,1,5): $\vec{\nabla} g = 8\hat{\imath} - 6\hat{\jmath} - \hat{k}$

Normal vector, $\overrightarrow{n_2} = \vec{\nabla} g = 8\hat{\imath} - 6\hat{\jmath} - \hat{k}$

Then angle between surfaces f and g is given by:

$$\cos\theta = [\widehat{n_1}.\widehat{n_2}] = \left[\frac{\overrightarrow{n_1}}{|\overrightarrow{n_1}|}.\frac{\overrightarrow{n_2}}{|\overrightarrow{n_2}|}\right]$$

$$\Rightarrow \theta = cos^{-1}\left[\frac{\vec{\nabla} f}{|\vec{\nabla} f|}.\frac{\vec{\nabla} g}{|\vec{\nabla} g|}\right] \qquad \Rightarrow \theta = cos^{-1}\left[\frac{(4\hat{\imath}+2\hat{\jmath}-\hat{k})}{\sqrt{21}}.\frac{(8\hat{\imath}-6\hat{\jmath}-\hat{k})}{\sqrt{101}}\right]$$

$$\Rightarrow \theta = cos^{-1}\left[\frac{32-12+1}{\sqrt{21}\sqrt{101}}\right] = cos^{-1}\left[\sqrt{\frac{21}{101}}\right]$$ **Answer.**